What explains the differences in birth trends across U.S. counties?

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ABSTRACT
Birth counts in the United States have declined over the past decade, yet the magnitude of this decline varies considerably across counties. This study investigates county-level birth counts from 2011 to 2020 to identify factors associated with differences in both baseline levels and temporal trends. The analysis evaluates the roles of population size, rural–urban classification (RUCC), socioeconomic deprivation (ADI), and demographic composition summarized through principal components. Generalized Estimating Equations (GEE) are used to account for repeated yearly measurements within counties, and alternative variance structures are considered to address overdispersion. Birth counts scale approximately proportionally with population size and exhibit a consistent downward temporal trend. Socioeconomic deprivation remains positively associated with birth counts after adjustment, while rurality primarily influences the rate of decline rather than baseline levels. Demographic structure provides additional explanatory value, capturing systematic heterogeneity across counties.

1 INTRODUCTION

Birth counts differ markedly across U.S. counties due to variation in population size, demographic composition, socioeconomic conditions, and settlement patterns. While national birth counts have declined in recent years, counties do not exhibit uniform trajectories. Some counties experience sharper declines, whereas others show more gradual changes. Understanding this heterogeneity is important for interpreting demographic dynamics at the local level.

This study examines whether differences in county-level birth trends between 2011 and 2020 are associated with structural county characteristics. The analysis focuses on three primary questions. First, do birth counts scale proportionally with population size (Figure 1)? Second, are rurality and socioeconomic deprivation associated with variation in birth counts after adjusting for population? Third, do these factors modify the rate at which birth counts change over time? In addition, demographic composition is incorporated through dimensionality reduction techniques to assess whether latent demographic structure explains residual variation beyond contextual predictors (Figure 5)

2 DATA AND VARIABLES

The analysis uses a longitudinal county-level dataset containing annual birth counts from 2011–2020. Each observation corresponds to a county–year combination, yielding 5,538 observations across 572 counties. Counties contribute repeated yearly measurements, motivating methods that account for within-county dependence.

The response variable is the annual number of births per county. Because birth counts scale with population size, county population enters the model through its logarithm as an offset. The validity of proportional scaling is evaluated empirically (Figure 1).

Socioeconomic context is represented using the Area Deprivation Index (ADI), standardized to mean zero and unit variance. Urbanicity is measured using the Rural–Urban Continuum Code (RUCC), centered for interpretability. Both variables are time-invariant within counties.

Temporal effects are modeled using calendar year, centered at its mean. Centering stabilizes estimation and clarifies interpretation of main and interaction effects.

Demographic composition is incorporated through a high-dimensional matrix describing county population structure across sex, race, ethnicity, and age categories. These variables are log-transformed and summarized using principal components derived via singular value decomposition. The retained components capture dominant demographic variation (Figure 5).

3 METHODS

3.1 Correlation structure and estimation framework

The repeated yearly measurements within counties induce within-county correlation that violates independence assumptions. To obtain valid uncertainty estimates while preserving population-averaged interpretations, the analysis employs Generalized Estimating Equations (GEE) with counties treated as clusters. An exchangeable working correlation structure is specified to account for within-county dependence.

3.2 Assessment of overdispersion

Birth counts exhibit substantial overdispersion relative to a Poisson variance model. This is assessed by examining the relationship between county-level mean and variance of birth counts across years. Regressing log variance on log mean yields a slope notably greater than one, indicating variance increases faster than predicted under a Poisson assumption. Residual diagnostics based on absolute Pearson residuals further reveal systematic heteroscedasticity under Poisson models (Figure 3). These findings motivate consideration of alternative variance structures.

3.3 Variance model comparison

Model specifications are compared using Poisson, Gamma, and Negative Binomial families. Poisson GEE models provide a baseline reference but show clear variance misfit. Gamma GEE models improve residual stability by allowing variance proportional to the square of the mean. Negative Binomial GEE models directly accommodate count overdispersion through an additional dispersion parameter. The Negative Binomial (Figure 4) specification is adopted as the primary model due to its alignment with the discrete nature of birth counts and improved variance behavior.

Population scaling is handled by including log population as an offset. This choice is supported by an empirical assessment of proportional scaling between log birth counts and log population, where the estimated slope approximates unity. Offset-based formulations therefore model relative birth intensity rather than absolute counts.

Temporal dynamics are modeled using centered calendar year. To assess heterogeneity in trends, interaction terms between year and both socioeconomic deprivation (ADI) and rural–urban classification (RUCC) are included. These interactions evaluate whether counties with differing structural characteristics exhibit distinct rates of change over time.

3.4 Demographic structure and PCR modeling

Demographic composition is incorporated through a principal components regression (PCR) framework. The high-dimensional demographic matrix is log-transformed and double-centered prior to singular value decomposition. Leading principal component (PC) scores are included as predictors to summarize dominant demographic variation while avoiding overparameterization. Models with varying numbers of PCs (0, 5, 10, 20, ...) are estimated to evaluate incremental explanatory contributions. Nested model comparisons are conducted using score tests, and variance-explained diagnostics (Figure 5) guide selection of retained components. To aid interpretation, PC coefficients are transformed back to the original demographic variable space. Nested comparisons are summarized in Table 3.

4 RESULTS

Birth counts scale approximately proportionally with county population size (Figure 1), supporting the offset-based modeling framework.

County-level birth counts exhibit overdispersion relative to a Poisson variance structure (Figure 2). Residual diagnostics confirm variance misfit under Poisson models (Figure 3), while the Negative Binomial model improves residual stability (Figure 4). Comparative model summaries are provided in Table 2.

Birth counts display a statistically robust declining temporal trend. Parameter estimates are summarized in Table 1.

Socioeconomic deprivation remains positively associated with birth counts, while RUCC does not show a statistically reliable main association (Table 1).

Both deprivation and rurality modify temporal dynamics (Table 1).

Incorporating demographic structure through principal components improves model fit. Variance-explained diagnostics are shown in Figure 5, and nested PCR comparisons are summarized in Table 3. Model fits under varying numbers of retained components are illustrated in Figures 6–8.

5 DISCUSSION AND FINDINGS

5.1 Population scaling and temporal dynamics

This analysis examined heterogeneity in county-level birth counts between 2011 and 2020, focusing on population scaling, socioeconomic deprivation, rurality, and demographic structure. Birth counts scale approximately proportionally with population size, supporting rate-based modeling through an offset formulation. After accounting for this scaling and within-county dependence, birth counts exhibit a consistent declining temporal pattern.

5.2 Socioeconomic deprivation and rurality

Socioeconomic deprivation remains a statistically stable predictor of birth counts. Counties with higher deprivation levels tend to have modestly higher expected births, even after adjustment for population size and clustering. Rural–urban classification does not show a reliable main association once deprivation is included, suggesting that deprivation captures much of the structural variation otherwise attributed to settlement patterns.

5.3 Heterogeneity in trends

Although rurality does not strongly differentiate baseline birth counts, it significantly modifies temporal trajectories. More rural counties experience steeper declines over time, while counties with higher deprivation display a less pronounced downward trend. These interaction patterns indicate that temporal changes in birth counts vary systematically across county contexts rather than following a uniform decline.

5.4 Role of demographic structure

Demographic composition provides substantial additional explanatory value. Principal components derived from the high-dimensional demographic structure capture dominant variation that improves model fit beyond socioeconomic and geographic predictors. This result indicates that birth count variation reflects structured demographic differences rather than residual noise.

County-level birth dynamics during this period are characterized by proportional population scaling, a widespread downward temporal trend, and systematic heterogeneity associated with deprivation, rurality, and demographic structure.

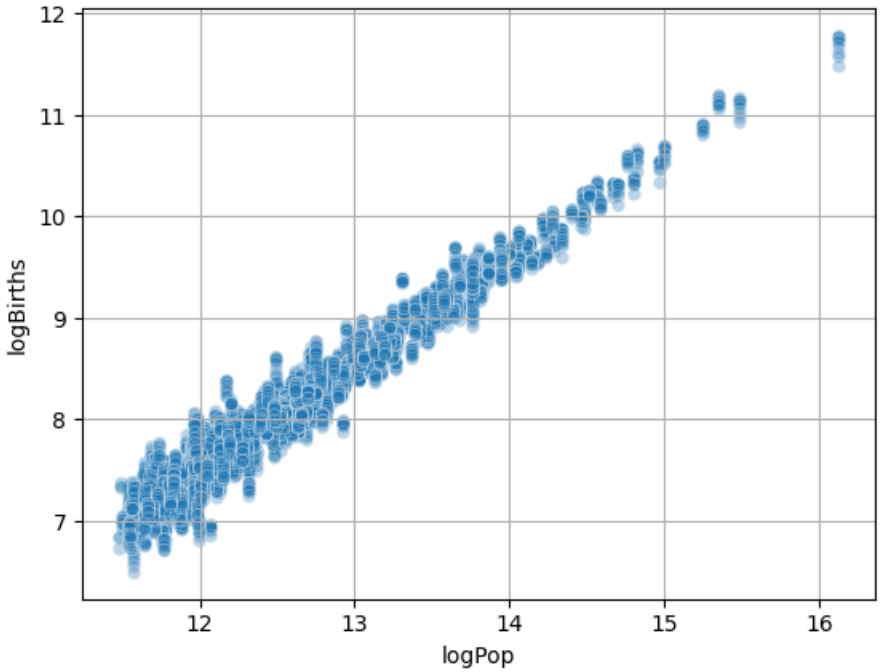

Figure 1. Relationship between county-level birth counts and population size on the logarithmic scale. The near-linear association with slope close to unity supports proportional scaling between births and population, motivating the use of log population as an offset in rate-based models.

Table 1. Negative Binomial / Gamma GEE model with time interactions

Variable	Coef	Std.Err	z	p-value
Intercept	-4.4382	0.007	-621.042	≤ 0.001
Deprivation (Z)	0.0346	0.007	4.653	≤ 0.001
RUCC (centered)	-0.0081	0.008	-0.959	0.338
Year (centered)	-0.0077	0.000	-17.143	≤ 0.001
Deprivation $\times$ Year	0.0011	0.000	2.182	0.029
RUCC $\times$ Year	-0.0032	0.001	-6.010	≤ 0.001

Table 2. Comparison of variance structures (GEE)

Model	Scale Parameter	Notes
Poisson GEE	130.954	Evidence of overdispersion
Gamma GEE	0.032	Improved variance stability
Negative Binomial GEE	0.031	Best residual behavior

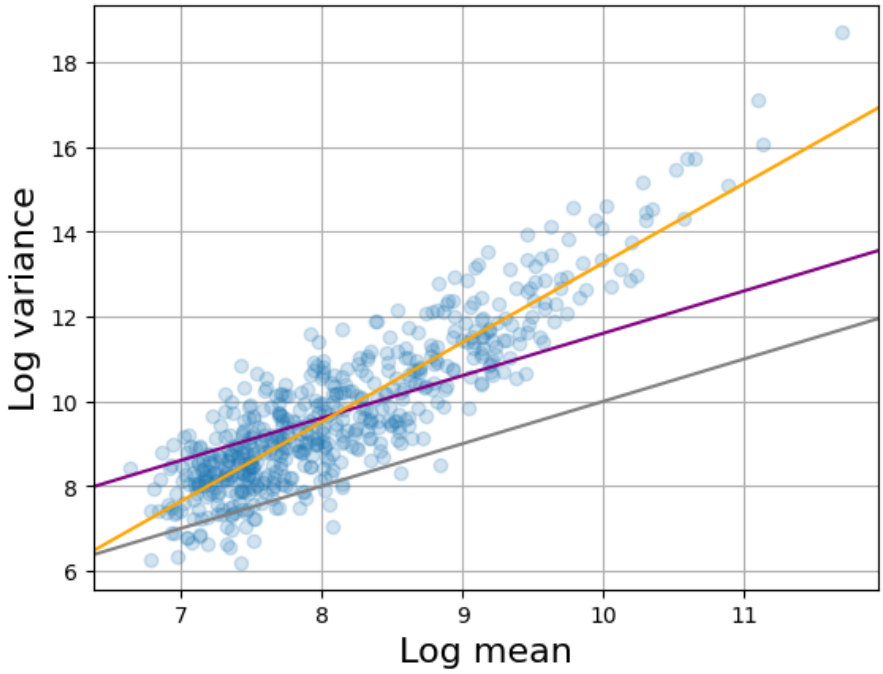

Figure 2. Empirical relationship between the logarithm of the variance and the logarithm of the mean of county-level birth counts across years. The estimated slope exceeding one indicates overdispersion relative to a Poisson variance structure.

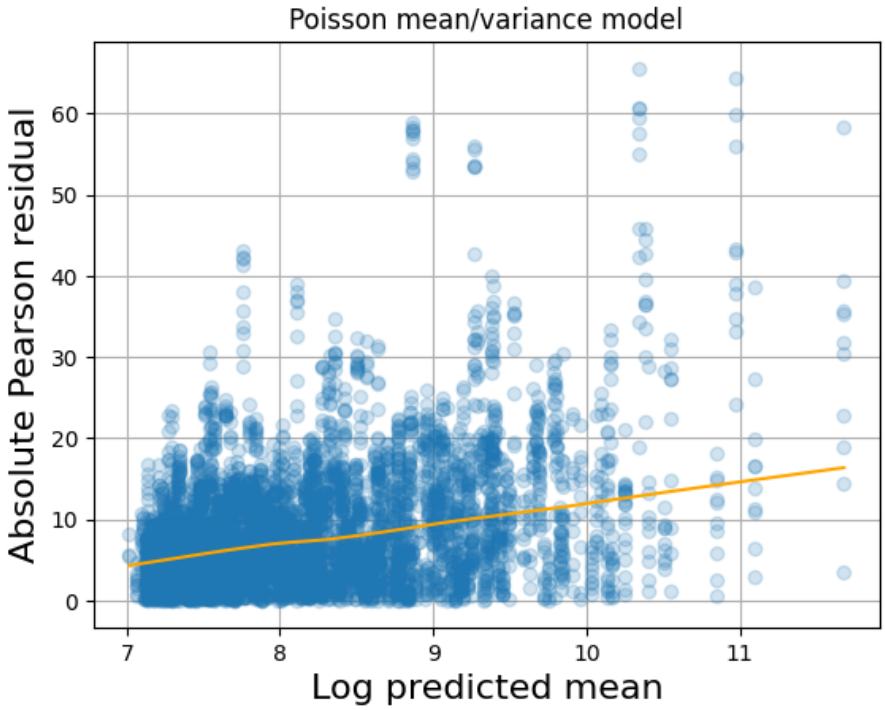

Figure 3. Absolute Pearson residuals versus fitted values under the Poisson GEE model. The systematic increase in residual spread with fitted values indicates variance misfit and motivates alternative variance specifications.

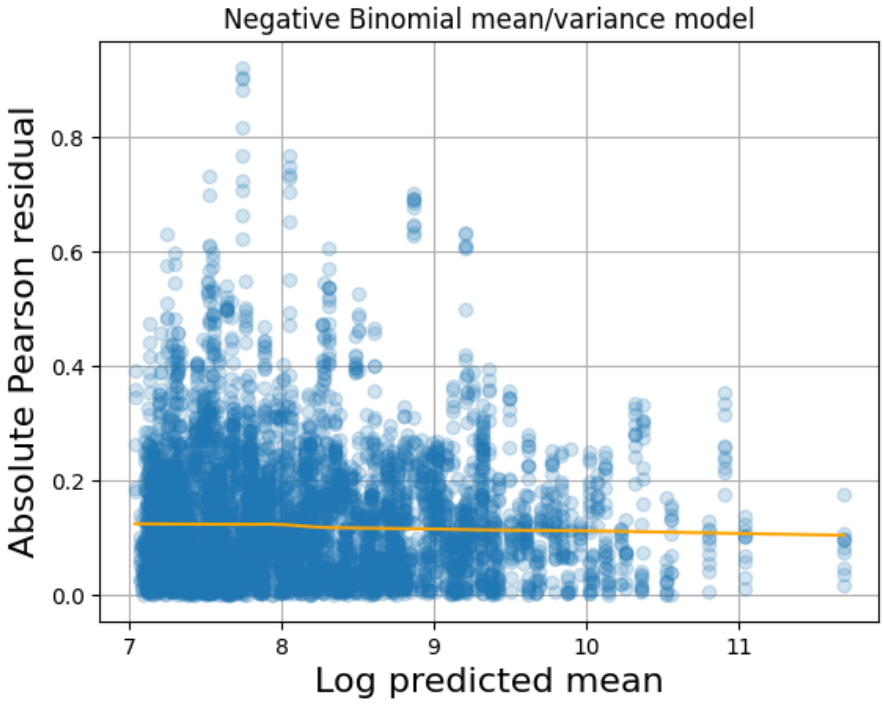

Figure 4. Absolute Pearson residuals versus fitted values under the Negative Binomial GEE model. Residual variability appears more stable across the fitted range, consistent with improved accommodation of overdispersion.

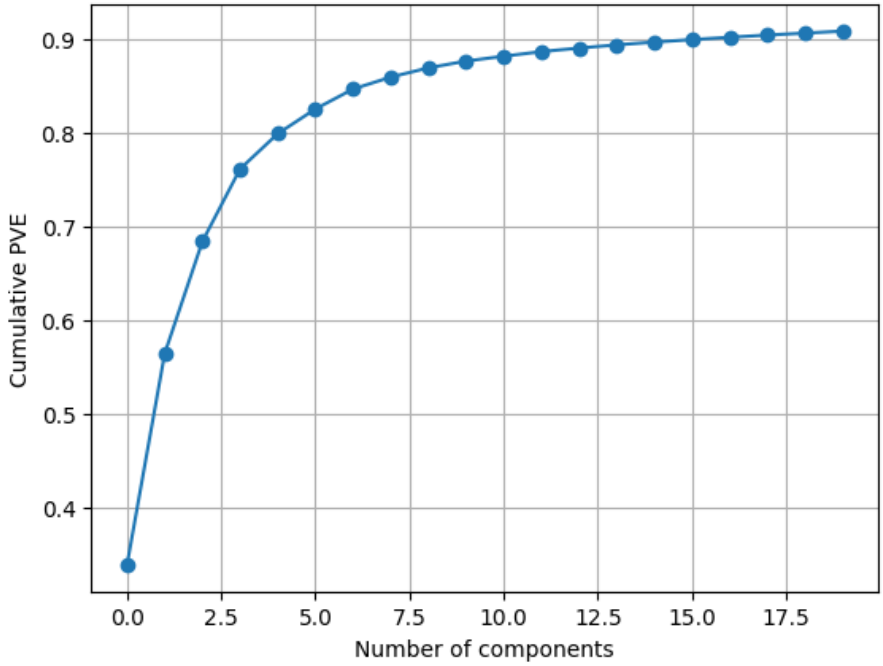

Figure 5. Cumulative proportion of variance explained (PVE) by leading principal components derived from the demographic composition matrix. Most variability concentrates within a limited number of dominant components.

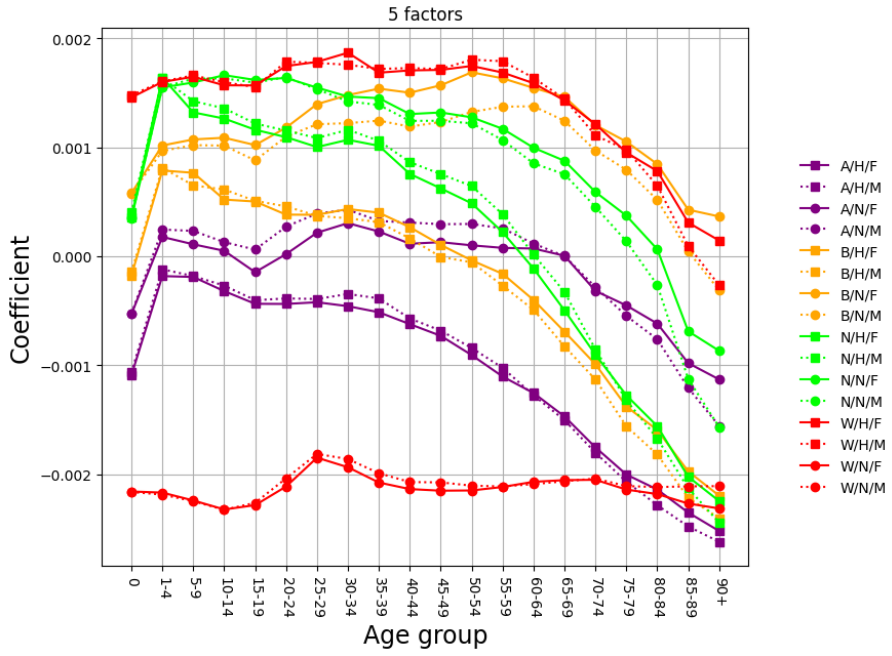

Figure 6. Model fit under principal components regression retaining five demographic factors.

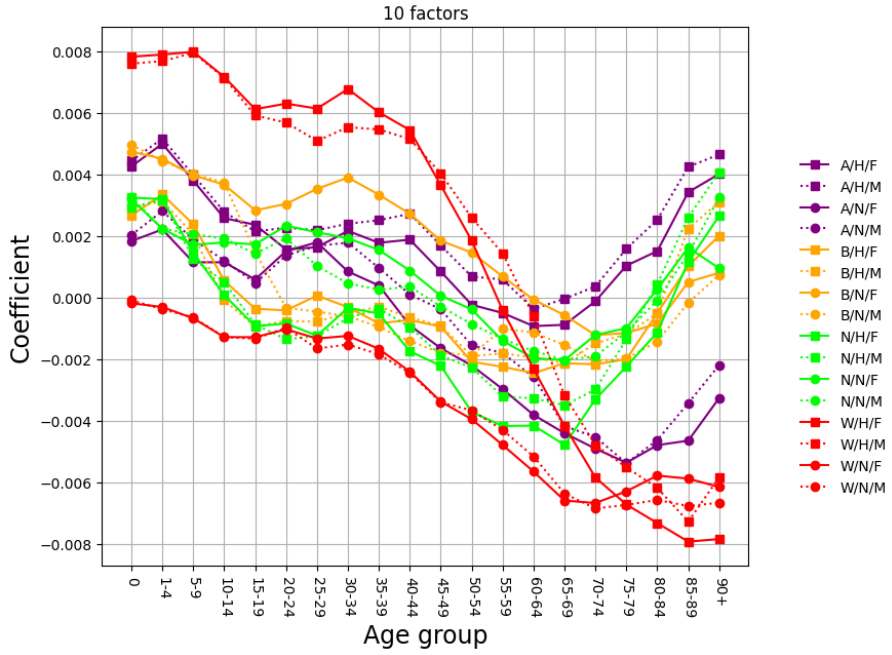

Figure 7. Model fit under principal components regression retaining ten demographic factors.

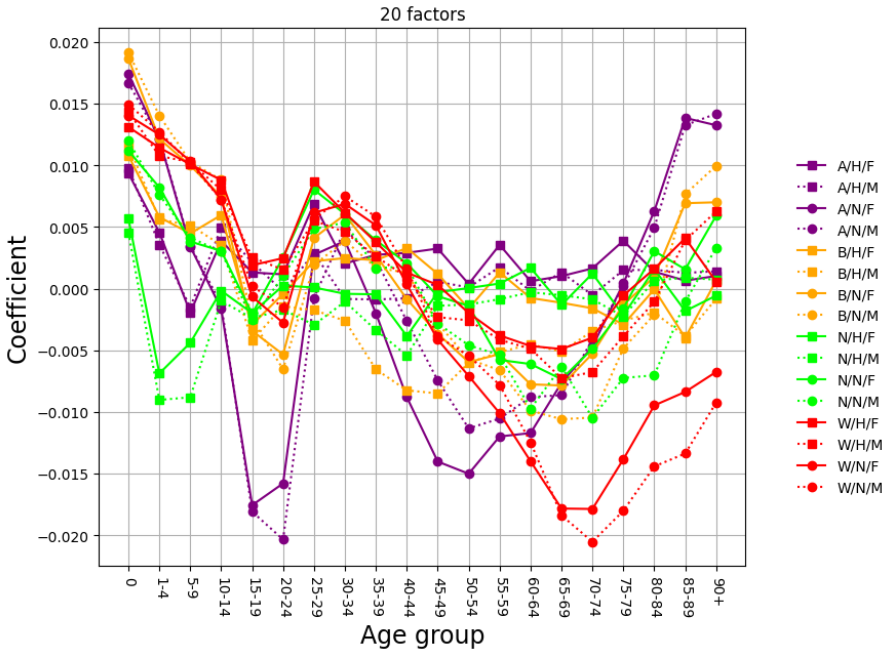

Figure 8. Model fit under principal components regression retaining twenty demographic factors.

Table 3. Score tests for nested principal components regression models

Comparison	p-value
5 vs 0 factors	≤ 0.001
10 vs 5 factors	≤ 0.001
20 vs 10 factors	≤ 0.001
30 vs 20 factors	≤ 0.001
40 vs 30 factors	0.000054
50 vs 40 factors	0.043915
60 vs 50 factors	0.000035
70 vs 60 factors	0.414524
80 vs 70 factors	0.032338
90 vs 80 factors	0.354696